

CATAMARCA, Argentina

WMO: 872220

Lat: 28.5936S

Lon: 65.7553W

Elev: 454

StdP: 95.99

Time Zone: -3.00 (W03)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	1.9	3.6	-8.0	2.0	13.9	-6.0	2.4	12.9	14.4	17.3	12.6	16.9	1.9	50	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	38.5	22.1	37.0	21.8	35.8	21.6	24.4	33.9	23.8	33.1	23.2	32.2	7.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.9	17.6	27.7	21.2	16.8	27.6	20.7	16.2	27.4	76.9	34.0	74.3	33.3	72.0	32.6	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	14.1	12.5	DB	-1.8	41.4	2.1	1.2	-3.3	42.3	-4.5	43.0	-5.7	43.6	-7.2	44.5	(4)
(5)			WB	-3.0	26.2	1.9	1.2	-4.3	27.1	-5.4	27.7	-6.5	28.4	-7.8	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	21.4	28.2	27.0	24.8	21.0	16.7	12.8	16.2	20.1	23.9	26.0	27.6	(6)	
(7)		DBStd	6.58	3.27	3.65	3.49	3.75	3.55	2.99	3.32	3.99	4.54	4.22	3.95	3.54	(7)
(8)		HDD10.0	25	0	0	0	0	1	8	12	3	1	0	0	0	(8)
(9)		HDD18.3	563	0	1	2	17	74	166	174	89	33	7	1	0	(9)
(10)		CDD10.0	4184	566	475	460	331	209	93	98	194	303	430	480	544	(10)
(11)		CDD18.3	1683	307	242	204	97	24	1	2	23	86	179	232	287	(11)
(12)		CDH23.3	19025	3804	2752	1890	756	176	18	51	352	943	1956	2750	3578	(12)
(13)	CDH26.7	9262	2037	1377	809	246	31	1	11	123	388	936	1394	1908	(13)	
(14)	Wind	WSAvg	5.1	6.1	5.8	5.3	4.2	3.8	2.9	3.5	4.4	5.6	6.4	6.7	6.6	(14)
(15)	Precipitation	PrecAvg	391	75	72	47	29	12	9	5	3	6	24	44	65	(15)
(16)		PrecMax	635	178	172	95	118	52	41	26	22	40	87	115	131	(16)
(17)		PrecMin	228	18	19	6	0	0	1	0	0	0	0	2	3	(17)
(18)		PrecStd	124	38	41	27	28	12	10	6	6	8	21	29	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.9	39.1	36.5	33.9	29.5	25.5	27.8	34.3	36.8	38.6	39.5	40.3	(19)
(20)			MCWB	23.2	23.1	22.8	21.4	19.0	16.1	16.3	18.0	19.3	20.5	21.2	22.1	(20)
(21)		2%	DB	37.9	36.9	34.6	31.3	27.1	23.0	24.2	30.0	33.2	36.0	36.9	38.0	(21)
(22)			MCWB	22.6	22.7	22.4	20.3	17.9	14.6	14.1	16.3	17.7	20.0	21.0	22.0	(22)
(23)		5%	DB	36.2	35.1	32.9	29.6	25.2	21.2	22.3	27.2	30.9	33.9	35.1	36.2	(23)
(24)			MCWB	22.4	22.5	22.0	19.8	17.0	13.3	13.2	15.2	17.0	19.4	20.4	21.8	(24)
(25)		10%	DB	34.6	33.3	31.0	27.8	23.4	19.7	20.4	24.7	28.7	31.8	33.4	34.5	(25)
(26)			MCWB	22.2	22.3	21.3	19.2	16.2	12.7	12.1	13.9	15.8	18.4	19.7	21.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	25.1	24.7	23.2	20.1	17.3	17.2	18.7	20.7	23.2	23.4	24.7	(27)
(28)			MCDWB	35.4	34.4	33.3	30.2	26.4	23.0	26.1	31.3	33.0	34.2	34.6	35.2	(28)
(29)		2%	WB	24.3	24.3	23.7	22.1	18.8	15.7	15.6	17.3	19.1	21.7	22.3	23.7	(29)
(30)			MCDWB	34.0	33.4	31.8	27.8	25.2	21.2	22.6	28.3	30.8	32.1	33.1	33.9	(30)
(31)		5%	WB	23.6	23.6	22.9	21.2	18.0	14.6	14.1	15.8	18.0	20.5	21.6	23.1	(31)
(32)			MCDWB	33.1	32.4	30.3	26.8	23.8	19.6	20.6	25.9	28.6	30.9	32.0	33.2	(32)
(33)		10%	WB	23.0	23.0	22.2	20.2	17.0	13.5	12.7	14.5	16.7	19.6	20.9	22.4	(33)
(34)			MCDWB	32.1	31.4	29.0	26.1	22.2	18.3	19.1	23.5	26.7	29.2	30.9	31.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	10.4	10.1	10.4	11.0	12.4	13.7	14.1	13.2	12.0	11.4	11.2	(35)
(36)			MCDBR	12.1	11.9	12.0	13.1	13.3	15.2	17.0	18.6	16.2	14.7	13.1	12.9	(36)
(37)			MCWBR	3.8	3.5	4.1	4.9	6.0	8.2	8.9	8.5	6.4	5.0	4.5	4.0	(37)
(38)		5% WB	MCDWB	11.1	10.8	10.6	10.7	11.7	13.2	15.2	17.7	14.6	13.0	11.8	11.5	(38)
(39)			MCWBR	4.0	3.8	4.2	4.5	5.6	7.5	8.3	8.4	6.2	4.8	4.3	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.401	0.394	0.375	0.361	0.327	0.314	0.311	0.350	0.400	0.409	0.399	0.400	(40)	
(41)		taud	2.409	2.430	2.482	2.466	2.510	2.516	2.498	2.373	2.235	2.289	2.356	2.397	(41)	
(42)		Ebn at Noon	942	931	917	880	867	857	876	875	875	906	937	945	(42)	
(43)		Edh at Noon	126	120	108	99	86	81	85	107	135	137	132	128	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.92	6.21	5.27	4.25	3.44	3.23	3.65	4.64	5.71	6.47	7.07	7.13	(44)	
(45)		RadStd	0.41	0.41	0.38	0.37	0.34	0.27	0.22	0.24	0.38	0.48	0.47	0.35	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+0.67	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page